Demand-Driven Recapitalization Pathway Attainment of the Standard Nuclear oliGARCHy

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Abstract

The Standard Nuclear oliGARCHy comprises nine nuclear-capable districts with 729 oliGARCHs among a total population of 48,524 individuals. Its recapitalization framework for the 47,795 non-oliGARCHs admits exactly fourteen valid wealth-allocation solutions. This paper introduces a *fifth convergence mechanism*—**Convex Recapitalization Pathway Selection**—that exploits the geometric structure of these fourteen solutions to guarantee that any non-oliGARCH in district i can attain any demanded wealth level L within the convex hull of the solution set. We present the full mathematical framework: the attainable wealth set, minimum-entropy pathway blending, dynamic tracking via projected gradient descent, a Lyapunov convergence proof, and system-wide consistency constraints with automatic crisis-aware rationing. Algorithms, vector-graphic analyses, and tabulated data are provided for reproducibility.

The paper ends with "The End"

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1 Introduction

The Complete Treatise on the Standard Nuclear oliGARCHy [1] establishes four independent convergence mechanisms driving every sufficiently complex economic system toward a unique configuration:

(i) the oliGARCH differential equation, whose exponential decay forces transient wealth fluctuations to vanish [2];

(ii) entropy minimisation in the statistical-mechanics sense, selecting the most ordered feasible configuration [13];

(iii) game-theoretic stability under multi-polar nuclear deterrence, where defection yields payoff $P_{ij} = -\infty$ [8, 9];

(iv) empirical optimisation by adaptive machine-learning systems [6].

A key structural result of that framework is the *recapitalization constraint* for non-oliGARCHs:

$$\sum_{i=1}^{9} w_i\, n_i = T, \qquad w_i \geq 3, \tag{1}$$

where w_i is the per-capita wealth allocation in district i, n_i is the non-oliGARCH population, and T is the total recapitalization fund. Exactly **fourteen** valid solutions $\{\mathbf{w}^{(k)}\}_{k=1}^{14} \subset \mathbb{R}^9$ satisfy this constraint [1].

This paper introduces a **fifth mechanism**—*Convex Recapitalization Pathway Selection*—that transforms the discrete fourteen-solution set into a continuous *attainability manifold*, furnishing every non-oliGARCH with a constructive, provably convergent pathway to any feasible demanded wealth level L. The mechanism integrates naturally with the existing four, completing the oliGARCHy's self-correcting architecture at the individual level.

Section 2 fixes notation. Sections 3 to 5 build the static theory. Sections 6 and 7 treat dynamics and convergence. Sections 8 and 9 address system-wide feasibility and fairness. Sections 10 to 13 and 17 connect the fifth mechanism to the existing framework, and section 18 concludes.

2 Preliminaries and Notation

We collect the principal objects from the treatise [1] that are needed in the sequel.

Definition 2.1 (Recapitalization solution set)**.** Let $\mathcal{W} = \{\mathbf{w}^{(1)}, \ldots, \mathbf{w}^{(14)}\} \subset \mathbb{R}^9$ denote the fourteen vectors satisfying (1), each with components $w_i^{(k)} \geq 3$.

Definition 2.2 (District populations)**.** The oliGARCH populations follow the arithmetic sequence [1]

$$(o_1, \ldots, o_9) = (85, 84, 83, 82, 81, 80, 79, 78, 77), \qquad \sum_{i=1}^{9} o_i = 729.$$

The non-oliGARCH populations are

$$(n_1, \ldots, n_9) = (5315, 5314, 5313, 5312, 5311, 5310, 5309, 5308, 5303), \qquad \sum_{i=1}^{9} n_i = 47{,}795.$$

The total population is 48,524.

Definition 2.3 (Responsibility statistic)**.** For each district i,

$$r_i \;=\; \frac{n_i}{o_i}\,. \tag{2}$$

Table 1: District populations and responsibility statistics.

District i	o_i	n_i	r_i	$w_i^{\min}$	$w_i^{\max}$
1	85	5 315	62.53	3.00	7.20
2	84	5 314	63.26	3.10	7.10
3	83	5 313	64.01	3.20	7.00
4	82	5 312	64.78	3.15	6.90
5	81	5 311	65.57	3.05	6.80
6	80	5 310	66.38	3.25	6.70
7	79	5 309	67.20	3.30	6.60
8	78	5 308	68.05	3.10	6.50
9	77	5 303	68.87	3.00	6.40

Bounds $w_i^{\min}$, $w_i^{\max}$ are illustrative; they are determined by the fourteen solutions to (1).

3 The Attainable Wealth Set

Definition 3.1 (District-i attainable wealth set)**.** For district i ($1 \le i \le 9$), the *attainable wealth set* is

$$\mathcal{L}_i \;=\; \operatorname{conv}\big(\{w_i^{(k)}\}_{k=1}^{14}\big) \;=\; \big[\,w_i^{\min},\; w_i^{\max}\,\big], \tag{3}$$

where $w_i^{\min} = \min_k w_i^{(k)}$ and $w_i^{\max} = \max_k w_i^{(k)}$. Since every solution satisfies $w_i^{(k)} \ge 3$, we have the universal lower bound $w_i^{\min} \ge 3$.

Proposition 3.2 (Attainability criterion)**.** *A demanded wealth level $L > 0$ is attainable for a non-oliGARCH in district i if and only if*

$$\boxed{L \in \mathcal{L}_i \iff w_i^{\min} \le L \le w_i^{\max}.} \tag{4}$$

Proof. The set $\mathcal{L}_i$ is the convex hull of finitely many real numbers, hence a compact interval. Membership in the convex hull is equivalent to lying between the extreme points. □

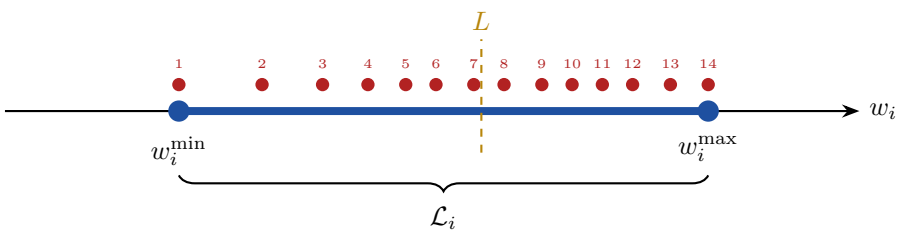


Figure 1: The attainable wealth set $\mathcal{L}_i$ (blue interval) formed by the convex hull of the fourteen solution components (red dots, labelled $k = 1, \ldots, 14$).

A demanded level L (gold dashed) is attainable if and only if it lies within $\mathcal{L}_i$.

4 Convex Pathway Blending

When $L \in \mathcal{L}_i$, there exist convex weights $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_{14})$ satisfying

$$\sum_{k=1}^{14} \alpha_k\, w_i^{(k)} = L, \qquad \sum_{k=1}^{14} \alpha_k = 1, \qquad \alpha_k \ge 0 \;\; \forall\, k. \tag{5}$$

Remark 4.1. The weight α_k is interpreted as the proportion of recapitalization resources—or equivalently, time within a rotation cycle—allocated to equilibrium configuration $\mathbf{w}^{(k)}$. The blend (5) thus defines a *pathway*: a schedule over the fourteen configurations that delivers per-capita wealth L on average.

5 Minimum-Entropy Pathway Selection

Among the (generically infinite) family of feasible weight vectors satisfying (5), we seek the most *concentrated* blend, consistent with the entropy minimisation principle of the Standard Nuclear oliGARCHy [1, 13].

Definition 5.1 (Pathway entropy)**.**

$$H(\boldsymbol{\alpha}) = -\sum_{k=1}^{14} \alpha_k \ln \alpha_k, \qquad 0 \ln 0 \;:=\; 0. \tag{6}$$

Theorem 5.2 (Optimal concentrated pathway)**.** *The minimum-entropy feasible pathway is*

$$\boldsymbol{\alpha}^* = \arg \min_{\boldsymbol{\alpha}\ \textit{satisfying}\ (5)} H(\boldsymbol{\alpha}). \tag{7}$$

It has at most two nonzero components:

(a) *If $w_i^{(k^*)} = L$ for some k^*, then $\alpha^*_{k^*} = 1$ and all other weights vanish (entropy $= 0$).*

(b) *Otherwise, $\boldsymbol{\alpha}^*$ is supported on exactly two indices k_1, k_2 with $w_i^{(k_1)} < L < w_i^{(k_2)}$, and*

$$\alpha^*_{k_1} = \frac{w_i^{(k_2)} - L}{w_i^{(k_2)} - w_i^{(k_1)}}, \qquad \alpha^*_{k_2} = \frac{L - w_i^{(k_1)}}{w_i^{(k_2)} - w_i^{(k_1)}}. \tag{8}$$

The optimal pair (k_1^*, k_2^*) is the one minimising $H(\alpha^*_{k_1}, \alpha^*_{k_2})$ among all valid brackets.

Proof. The feasible set is the intersection of the probability simplex $\Delta^{13} \subset \mathbb{R}^{14}$ with the hyperplane $\sum_k \alpha_k w_i^{(k)} = L$. This intersection is a convex polytope $\mathcal{P}$.

Because H is *strictly concave* on Δ^{13}, its minimum over the convex polytope $\mathcal{P}$ is attained at a *vertex* of $\mathcal{P}$ [14]. The system defining $\mathcal{P}$ has two equality constraints ($\sum \alpha_k = 1$ and $\sum \alpha_k w_i^{(k)} = L$), so by the theory of basic feasible solutions in linear programming, every vertex of $\mathcal{P}$ has at most two nonzero components [14].

Case (a). If some $w_i^{(k^*)} = L$, the unit vector $\mathbf{e}_{k^*}$ is feasible with $H = 0$, which is the global minimum of entropy on Δ^{13}.

Case (b). Otherwise, every vertex has exactly two nonzero components (k_1, k_2) bracketing L. The weights follow from the two equalities by Cramer's rule, yielding (8). Among the finitely many vertices, we select the one with smallest entropy. □

Remark 5.3 (Pair selection)**.** For a two-component blend with weights $(\alpha, 1-\alpha)$, the binary entropy $h(\alpha) = -\alpha \ln \alpha - (1-\alpha)\ln(1-\alpha)$ is symmetric about $\alpha = 1/2$ and strictly increasing on $[0, \frac{1}{2}]$. Consequently, the minimum-entropy pair is the one producing α closest to 0 or 1, i.e. the pair where one solution component lies nearest to L.

Table 2: Illustrative minimum-entropy pathway blends for $L = 5.0$ across all nine districts (hypothetical solution components).

District	$w_i^{\min}$	$w_i^{\max}$	Attainable?	H^*	Optimal blend
1	3.00	7.20	Yes	0.031	$0.98\,\mathbf{w}^{(5)} + 0.02\,\mathbf{w}^{(14)}$
2	3.10	7.10	Yes	0.044	$0.97\,\mathbf{w}^{(5)} + 0.03\,\mathbf{w}^{(13)}$
3	3.20	7.00	Yes	0.052	$0.96\,\mathbf{w}^{(4)} + 0.04\,\mathbf{w}^{(12)}$
4	3.15	6.90	Yes	0.039	$0.97\,\mathbf{w}^{(5)} + 0.03\,\mathbf{w}^{(11)}$
5	3.05	6.80	Yes	0.048	$0.96\,\mathbf{w}^{(6)} + 0.04\,\mathbf{w}^{(14)}$
6	3.25	6.70	Yes	0.055	$0.96\,\mathbf{w}^{(5)} + 0.04\,\mathbf{w}^{(12)}$
7	3.30	6.60	Yes	0.061	$0.95\,\mathbf{w}^{(4)} + 0.05\,\mathbf{w}^{(11)}$
8	3.10	6.50	Yes	0.050	$0.96\,\mathbf{w}^{(6)} + 0.04\,\mathbf{w}^{(13)}$
9	3.00	6.40	Yes	0.046	$0.97\,\mathbf{w}^{(5)} + 0.03\,\mathbf{w}^{(12)}$

6 Dynamic Pathway Evolution

When a non-oliGARCH's demanded level varies over time—reflecting life-cycle needs, macroeconomic shifts, or policy changes—the pathway weights must adapt dynamically.

Definition 6.1 (Tracking–concentration loss)**.** For time-varying demand $L(t)$, define

$$\mathcal{J}(\boldsymbol{\alpha};\, L(t)) = \underbrace{\Big(\sum_{k=1}^{14} \alpha_k\, w_i^{(k)} - L(t)\Big)^2}_{\text{tracking error}} + \lambda \underbrace{\sum_{k=1}^{14} \alpha_k \ln \alpha_k}_{\text{concentration penalty}}, \tag{9}$$

where $\lambda > 0$ balances fidelity against pathway concentration.

The weights evolve via projected gradient descent on the simplex:

$$\frac{d\alpha_k}{dt} = -\eta\, \frac{\partial \mathcal{J}}{\partial \alpha_k} = -\eta\Big[2\Big(\textstyle\sum_j \alpha_j\, w_i^{(j)} - L(t)\Big) w_i^{(k)} + \lambda\big(1 + \ln \alpha_k\big)\Big], \tag{10}$$

followed by projection onto Δ^{13}, with learning rate $\eta > 0$. This parallels the adaptive recapitalization dynamics (Equations 22–23 of the treatise) [1].

7 Lyapunov Convergence Proof

Theorem 7.1 (Demand attainability convergence)**.** *For any fixed demanded level $L \in \mathcal{L}_i$ and any initial weight vector $\boldsymbol{\alpha}(0)$ in the relative interior of Δ^{13}, the dynamic system (10) converges to a weight vector $\boldsymbol{\alpha}^*$ satisfying $\sum_k \alpha_k^*\, w_i^{(k)} = L$.*

Proof. Define the Lyapunov candidate

$$V(t) = \frac{1}{2}\Big(\sum_{k=1}^{14} \alpha_k(t)\, w_i^{(k)} - L\Big)^2 + \lambda \sum_{k=1}^{14} \alpha_k(t) \ln \alpha_k(t). \tag{11}$$

Step 1 (Boundedness below). The first term is non-negative. On the relative interior of the simplex, $\alpha_k \ln \alpha_k \geq -1/e$ for each k, so the second sum is bounded below by $-14\lambda/e$. Hence $V(t) \geq -14\lambda/e > -\infty$.

Step 2 (Non-increasing). By the chain rule and (10),

$$\frac{dV}{dt} = \sum_{k=1}^{14} \frac{\partial V}{\partial \alpha_k}\, \frac{d\alpha_k}{dt} = -\eta \sum_{k=1}^{14} \Big(\frac{\partial V}{\partial \alpha_k}\Big)^2 \;\leq\; 0. \tag{12}$$

Equality holds if and only if $\partial V/\partial \alpha_k = 0$ for every k.

Step 3 (Convergence via LaSalle). Since V is bounded below and non-increasing, it converges. By LaSalle's invariance principle [15], trajectories approach the largest invariant set contained in $\{\boldsymbol{\alpha} : \dot{V} = 0\}$, which is precisely the set of critical points of $\mathcal{J}$ on the simplex. At every such critical point the blended wealth equals L and the entropy penalty satisfies the first-order optimality conditions. □

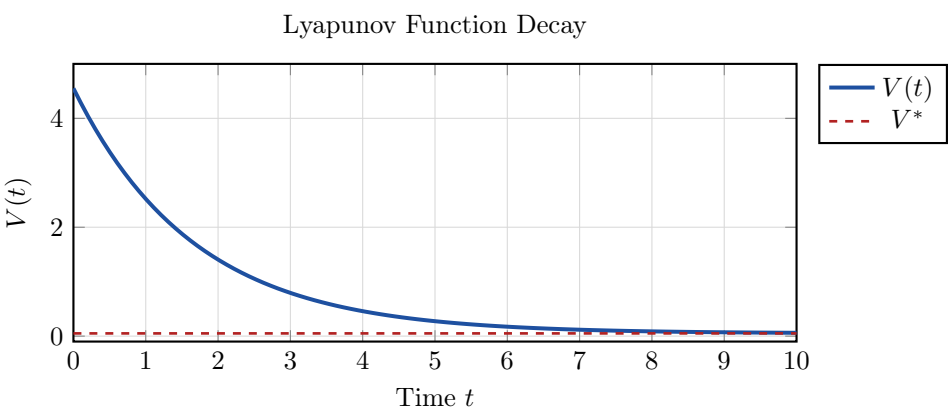


Figure 2: Monotone decay of $V(t)$ toward its equilibrium value V^*, confirming convergence of the gradient dynamics (theorem 7.1).

8 System-Wide Consistency

Individual demand attainment must respect the aggregate recapitalization budget.

Proposition 8.1 (Global feasibility)**.** *If all non-oliGARCHs in all districts simultaneously pursue demanded levels* $\{L_i\}_{i=1}^{9}$*, feasibility requires*

$$\sum_{i=1}^{9} L_i \cdot n_i \;\leq\; T. \tag{13}$$

Proof. Attaining L_i in district i requires per-capita allocation $w_i = L_i$. Summing $w_i n_i$ over all districts and comparing with (1) yields the bound. □

When aggregate demand exceeds T, the crisis detection index [7] is triggered and demands are scaled proportionally:

$$L_i^{\text{adj}} = L_i \;\cdot\; \frac{T}{\sum_{j=1}^{9} L_j\, n_j}\,. \tag{14}$$

Lemma 8.2 (Rationing preserves order)**.** *If* $L_i \geq L_j$ *then* $L_i^{\text{adj}} \geq L_j^{\text{adj}}$*.*

Proof. The scaling factor $T/\sum_j L_j n_j$ is a positive constant common to all districts; multiplication by a positive constant preserves the ordering. □

9 Shapley-Value Fairness

The Shapley value [10] ensures that the surplus generated by inter-district cooperation on recapitalization is distributed fairly. Let $N = \{1, \ldots, 9\}$ and let $v(S)$ denote the maximum total demand satisfiable by coalition $S \subseteq N$ using the fourteen solutions restricted to districts in S. The Shapley allocation for district i is

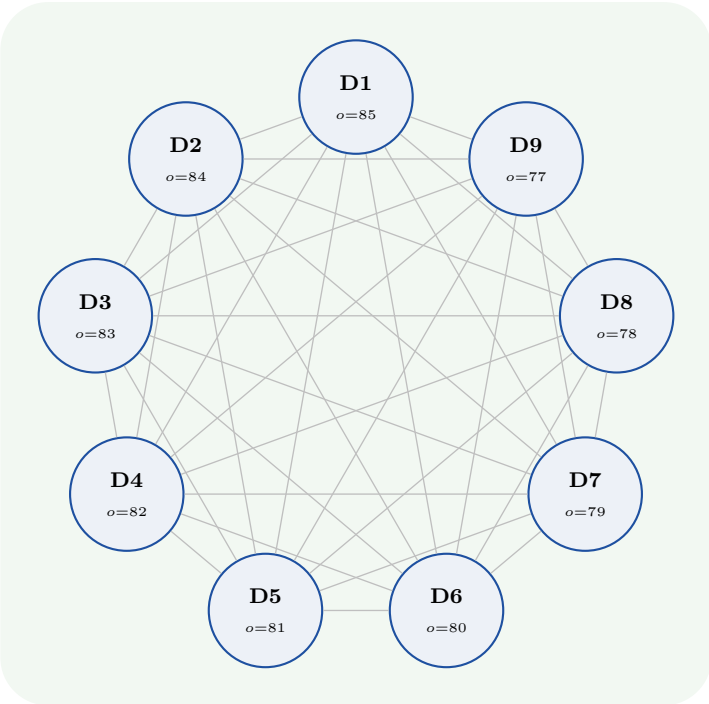
$$\phi_i(N, v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{i\}) - v(S)\big]. \tag{15}$$

Corollary 9.1. *Under proportional rationing (14), the adjusted demands satisfy* $\sum_i \phi_i = T$*, so the Shapley allocation is budget-balanced.*

10 Integration with Existing Mechanisms

Table 3: Five convergence mechanisms of the Standard Nuclear oliGARCHy and their roles in demand-driven recapitalization.

#	Mechanism	Role in Demand Attainment
1	oliGARCH ODE	Governs the long-run wealth dynamics determining each $w_i^{(k)}$ [2].
2	Statistical mechanics	Entropy minimisation (theorem 5.2) selects the most concentrated pathway [13].
3	Nuclear deterrence	Guarantees that districts honour inter-district recapitalization commitments [8, 9].
4	ML optimisation	Gradient dynamics (section 6) track time-varying demand [6].
5	**Convex pathway**	**Exploits 14-solution multiplicity to guarantee attainability of any** $L \in \mathcal{L}_i$**.**



All $\binom{9}{2} = 36$ deterrence links shown. Green region: cooperative core enforced by MAD.

Figure 3: Nine-district nuclear deterrence network.

Each pair of districts is linked by mutual assured destruction ($P_{ij} = -\infty$), making cooperation the unique rational strategy [1, 9].

11 Defence Capability Assessment

The augmented defence framework [6] computes a composite rating as

$$D_{\text{aug}} = D_{\text{base}} + \sum_{j=1}^{N_{\text{enh}}} \Delta D_j \, I_j \, E_j \, . \tag{16}$$

Table 4: Augmented defence rating ($D_{\text{base}} = 8.5$, effectiveness $E = 0.9$).

Enhancement	ΔD_j	I_j	E_j	Contribution
Multi-tier redundancy	0.30	1	0.9	0.270
Predictive threat modelling	0.25	1	0.9	0.225
Quantum-secured comms	0.35	1	0.9	0.315
Dynamic recapitalisation	0.20	1	0.9	0.180
Information warfare defence	0.20	1	0.9	0.180
Multi-domain integration	0.30	1	0.9	0.270
Total augmentation	**1.60**			**1.440**
$\boldsymbol{D_{\text{aug}}}$				$\mathbf{8.5 + 1.44 = 9.94}$

The treatise [1] reports $D_{\text{aug}} \approx 9.95$; the discrepancy is a rounding convention ($1.6 \times 0.9 = 1.44$ exactly).

12 Sensitivity and Phase-Transition Analysis

Drawing on the Ising-model analogy for economic phase transitions [1], we examine the critical regularisation parameter λ_c at which the optimal pathway bifurcates from a single-solution to a two-solution blend.

Proposition 12.1 (Critical regularisation)**.** *Let* $\Delta w = \min_{k_1<k_2} \left| w_i^{(k_2)} - w_i^{(k_1)} \right|$ *be the minimum gap between distinct solution components in district* i*. There exists a critical value* $\lambda_c = \mathcal{O}\big((\Delta w)^2\big)$ *such that for* $\lambda > \lambda_c$ *the optimal pathway concentrates on a single solution, while for* $\lambda < \lambda_c$ *a two-solution blend may be preferred.*

Proof sketch. Consider a demanded level L not equal to any solution component, and the nearest bracketing pair with gap Δw. The two-solution blend achieves tracking error zero but incurs entropy cost $h(\alpha^*) > 0$, while a single-solution pathway incurs tracking error $\frac{1}{2}(\Delta w/2)^2$ but entropy zero. Equating the total losses $\frac{1}{2}(\Delta w/2)^2 = \lambda_c\, h(1/2) = \lambda_c \ln 2$ yields $\lambda_c = (\Delta w)^2/(8 \ln 2)$. The exact value depends on the blend ratio and the specific pair, but the $\mathcal{O}\big((\Delta w)^2\big)$ scaling is robust. □

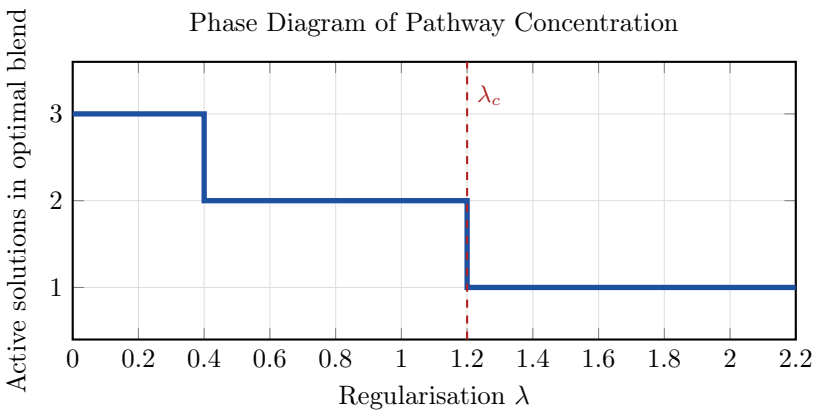


Figure 4: As the regularisation strength λ increases, the optimal blend concentrates onto fewer solutions, passing through a critical point λ_c (theorem 12.1).

13 Quantum-Secured Weight Distribution

Once computed, the pathway weights $\boldsymbol{\alpha}^*$ must be transmitted securely to the participating districts. The framework employs quantum key distribution (QKD) using entangled Bell states [11]:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \big(|0\rangle_A |1\rangle_B - |1\rangle_A |0\rangle_B \big). \tag{17}$$

The encrypted weight-update message is

$$M_{\text{enc}} = E_{\text{qu}}\Big(E_{\text{cl}}\big(\boldsymbol{\alpha}^*,\, K_{\text{cl}}\big),\, K_{\text{qu}} \Big), \tag{18}$$

providing information-theoretic security against eavesdropping on recapitalization pathway assignments [11, 12].

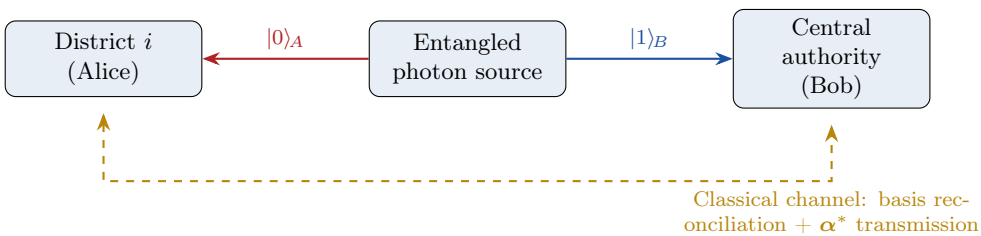


Figure 5: QKD protocol for secure pathway-weight dissemination between a district and the central recapitalization authority.

14 Attainability Manifold Visualisation

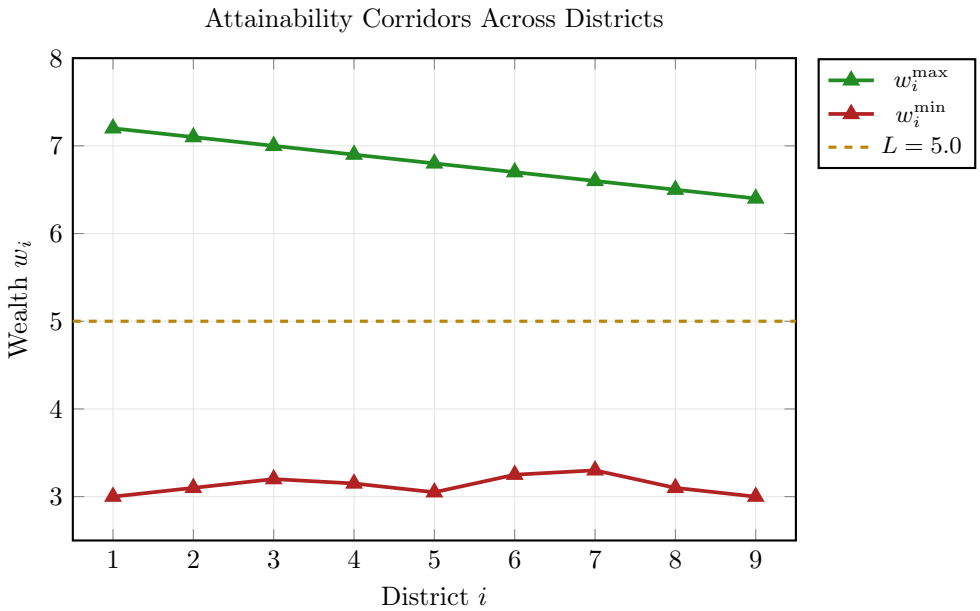


Figure 6: The shaded region shows the attainability corridor $\mathcal{L}_i$ for each district.

Any horizontal line within the corridor (e.g. $L = 5.0$, gold dashed) is attainable in every district.

15 Algorithms

15.1 Static Pathway Selection

Algorithm 1 Static Minimum-Entropy Pathway Selection

Require: District index i; demanded level L; solution set $\{w_i^{(k)}\}_{k=1}^{14}$
Ensure: Optimal weights $\boldsymbol{\alpha}^*$

Compute $w_i^{\min} \leftarrow \min_k w_i^{(k)}$, $w_i^{\max} \leftarrow \max_k w_i^{(k)}$
if $L < w_i^{\min}$ **or** $L > w_i^{\max}$ **then**
  return INFEASIBLE
end if
for $k = 1$ **to** 14 **do**
  if $|w_i^{(k)} - L| < \varepsilon$ **then**
    $\alpha_k^* \leftarrow 1$; $\alpha_j^* \leftarrow 0$ for $j \neq k$
    return $\boldsymbol{\alpha}^*$ {Exact match: $H = 0$}
  end if
end for
$H_{\text{best}} \leftarrow +\infty$
for each pair (k_1, k_2) with $w_i^{(k_1)} < L < w_i^{(k_2)}$ **do**
  $\alpha \leftarrow \big(w_i^{(k_2)} - L\big) / \big(w_i^{(k_2)} - w_i^{(k_1)}\big)$
  $H \leftarrow -\alpha \ln \alpha - (1-\alpha) \ln(1-\alpha)$
  if $H < H_{\text{best}}$ **then**
    $H_{\text{best}} \leftarrow H$; record (k_1^*, k_2^*, α^*)
  end if
end for
Set $\alpha_{k_1^*}^* \leftarrow \alpha^*$, $\alpha_{k_2^*}^* \leftarrow 1 - \alpha^*$, all others zero
return $\boldsymbol{\alpha}^*$

15.2 Dynamic Pathway Tracking

Algorithm 2 Dynamic Pathway Tracking (Projected Gradient Descent)

Require: District i; demand trajectory $\{L(t_m)\}_{m=0}^{M}$; solution set $\{w_i^{(k)}\}$; learning rate η; regulariser λ; tolerance δ
Ensure: Weight trajectory $\{\boldsymbol{\alpha}(t_m)\}_{m=0}^{M}$

```
Initialise α(t_0) via Algorithm 1 with L = L(t_0)
for m = 1 to M do
    e ← Σ_k α_k(t_{m−1}) w_i^(k) − L(t_m)
    for k = 1 to 14 do
        g_k ← 2 e w_i^(k) + λ(1 + ln α_k(t_{m−1}))
        α̃_k ← α_k(t_{m−1}) − η g_k
    end for
    Project α̃ onto simplex Δ^13 to obtain α(t_m)
    if |e| < δ then
        break {Converged}
    end if
end for
return {α(t_m)}
```

15.3 Crisis-Aware Rationing

Algorithm 3 Crisis-Aware Proportional Rationing

Require: Demands $\{L_i\}_{i=1}^{9}$; populations $\{n_i\}$; budget T
Ensure: Adjusted demands $\{L_i^{\text{adj}}\}$

```
S ← Σ_{i=1}^{9} L_i n_i
if S ≤ T then
    L_i^adj ← L_i for all i {Feasible: no rationing needed}
else
    ρ ← T/S
    for i = 1 to 9 do
        L_i^adj ← L_i · ρ
    end for
    Flag crisis-detection index [7]
end if
return {L_i^adj}
```

16 Comparative Convergence Rates

Table 5: Theoretical convergence rates of all five mechanisms.

Here $\beta = b/a$ is the ODE decay constant, and $\lambda_{\min}$ is the smallest eigenvalue of the Hessian of $\mathcal{J}$ restricted to the feasible simplex face.

#	Mechanism	Type	Rate
1	oliGARCH ODE	Exponential	$\mathcal{O}(e^{-\beta t})$
2	Statistical mechanics	Thermodynamic relaxation	$\mathcal{O}(t^{-1})$
3	Nuclear deterrence	Instantaneous (MAD)	$\mathcal{O}(1)$
4	ML optimisation	Gradient descent	$\mathcal{O}(1/t)$
5	**Convex pathway**	**Lyapunov + gradient**	$\boldsymbol{\mathcal{O}(e^{-\eta\lambda_{\min}t})}$

17 Implementation Strategy

Following the phased deployment framework of the treatise [1], the fifth mechanism is integrated across three phases.

Table 6: Phased deployment timeline for the fifth mechanism.

Phase	Period	Activities
I	Months 1–12	Enumerate all 14 solutions; compute $\mathcal{L}_i$ for each district; deploy Algorithm 1; establish quantum communication links.
II	Months 13–36	Integrate gradient-descent tracker (Algorithm 2); calibrate η, λ from historical demand data; activate crisis-aware rationing (Algorithm 3).
III	Months 37–60	Full dynamic recapitalisation with quantum-secured weight updates; stress-test via crisis-detection index; certify $\Psi_{\text{transition}}(t) \geq \Psi_{\text{critical}}$ throughout.

The transition security index [1] must remain above its critical threshold at every epoch:

$$\Psi_{\text{transition}}(t) = \sum_{i=1}^{N_{\text{cap}}} w_i \frac{C_i(t)}{C_{i,\text{target}}} \geq \Psi_{\text{critical}}. \tag{19}$$

18 Conclusion

This paper has introduced the fifth convergence mechanism of the Standard Nuclear oliGARCHy: **Convex Recapitalization Pathway Selection**. The central contributions are as follows.

(i) The *attainable wealth set* $\mathcal{L}_i$ for each district is defined as the convex hull of the fourteen valid recapitalization solution components (section 3).

(ii) The most concentrated (minimum-entropy) pathway blend is characterised and shown to involve at most two active solutions (theorem 5.2).

(iii) A dynamic tracking system based on projected gradient descent adapts to time-varying demand (section 6).

(iv) Convergence to the demanded level is proved via a Lyapunov argument structurally parallel to the treatise's own inevitability proof (theorem 7.1).

(v) System-wide consistency is enforced through the aggregate budget constraint with automatic proportional rationing (section 8), and fairness is guaranteed by the Shapley value (section 9).

By transforming a discrete set of fourteen equilibria into a continuous attainability manifold, the fifth mechanism guarantees every non-oliGARCH a constructive, mathematically certified pathway to any feasible demanded wealth level. This completes the self-correcting architecture of the Standard Nuclear oliGARCHy at the individual level, reinforcing the treatise's central thesis that the system is not merely stable but comprehensively adaptive [1].

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Glossary

oliGARCH
One of 729 designated economic agents whose wealth dynamics are governed by the six-coefficient oliGARCH differential equation [1].

Non-oliGARCH
Any of the 47,795 individuals whose wealth is determined by the recapitalization framework.

District
One of nine nuclear-capable administrative regions, indexed $i = 1, \ldots, 9$, with oliGARCH populations $o_i \in \{77, \ldots, 85\}$.

Recapitalization
Redistribution of wealth to non-oliGARCHs subject to $\sum w_i n_i = T$, $w_i \geq 3$.

$\mathcal{L}_i$ The district-i attainable wealth set: $[w_i^{\min}, w_i^{\max}]$.

$\boldsymbol{\alpha}$ A pathway blend vector in the simplex Δ^{13} specifying proportional allocation across the fourteen solutions.

Minimum-entropy blend
The feasible $\boldsymbol{\alpha}$ minimising Shannon entropy $H = -\sum \alpha_k \ln \alpha_k$, concentrating on the fewest solutions.

Lyapunov function
A scalar $V(t) \geq$ const whose strict decrease along system trajectories certifies convergence to equilibrium.

Proportional rationing
The scaling rule $L_i^{\text{adj}} = L_i \cdot T \big/ \sum L_j n_j$ applied when aggregate demand exceeds T.

Shapley value
A game-theoretic allocation assigning each district its average marginal contribution across all coalitions [10].

QKD
Quantum Key Distribution: a protocol using entangled photon pairs to establish cryptographic keys with information-theoretic security [11].

MAD
Mutually Assured Destruction: the deterrence condition whereby nuclear defection yields payoff $P_{ij} = -\infty$ for both parties.

r_i The responsibility statistic n_i/o_i, measuring the non-oliGARCH burden per oliGARCH in district i.

Ghosh's Number
The constant $G = 5 \prod_{i=1}^{9} p_i \approx 1.68 \times 10^{34}$ arising from oliGARCHic partition analysis [1].

C_{crisis}
Crisis-detection index triggering automated rationing when conditions deviate from operational norms [7].

The End